

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce **Department of Information Technology (B.Sc.I.T Semester V)**
Applied Business Analytics

Practical – 2

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Date of Assignment: 31/07/26	Date/Time of Submission: 31/07/26

Aim:-

2. Describing the Distribution of a Variable:

a. Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest.

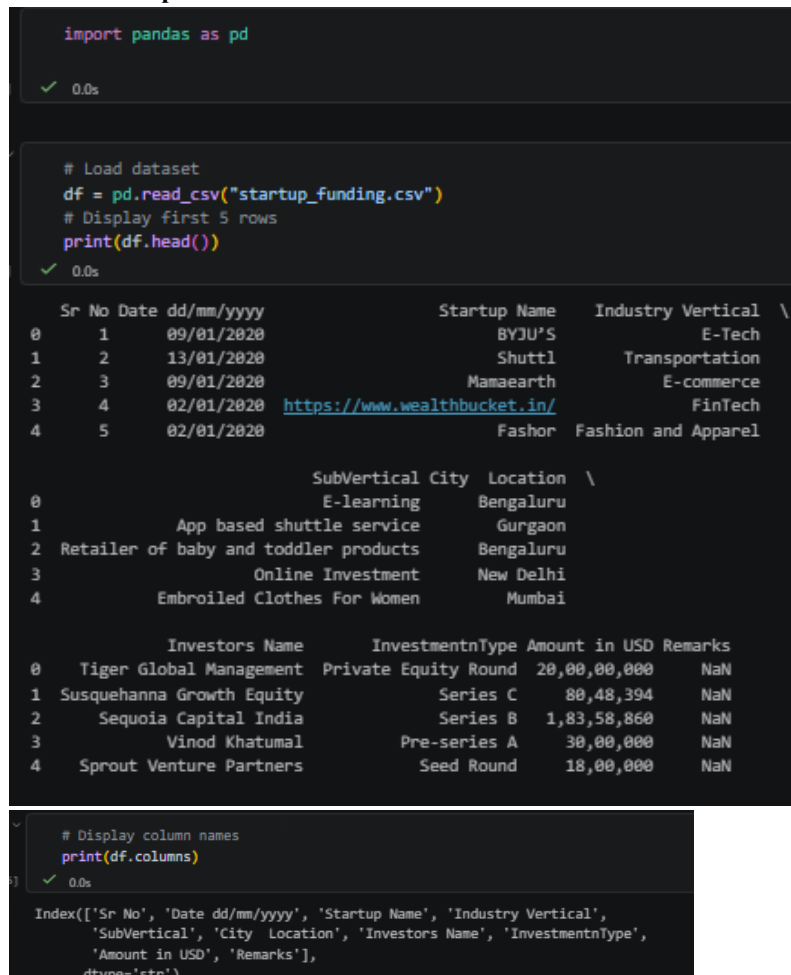
Code and output:-

```
import pandas as pd

# Load dataset
df = pd.read_csv("startup_funding.csv")
# Display first 5 rows
print(df.head())

# Display column names
print(df.columns)

Index(['Sr No', 'Date dd/mm/yyyy', 'Startup Name', 'Industry Vertical',
       'SubVertical', 'City Location', 'Investors Name', 'InvestmentType',
       'Amount in USD', 'Remarks'],
      dtype='str')
```



Sr No	Date dd/mm/yyyy	Startup Name	Industry Vertical	SubVertical	City Location
0	09/01/2020	BYJU'S	E-Tech	E-learning	Bengaluru
1	13/01/2020	Shuttl	Transportation	App based shuttle service	Gurgaon
2	09/01/2020	Mamaearth	E-commerce	Retailer of baby and toddler products	Bengaluru
3	02/01/2020	https://www.wealthbucket.in/	FinTech	Online Investment	New Delhi
4	02/01/2020	Fashor	Fashion and Apparel	Embroided Clothes For Women	Mumbai

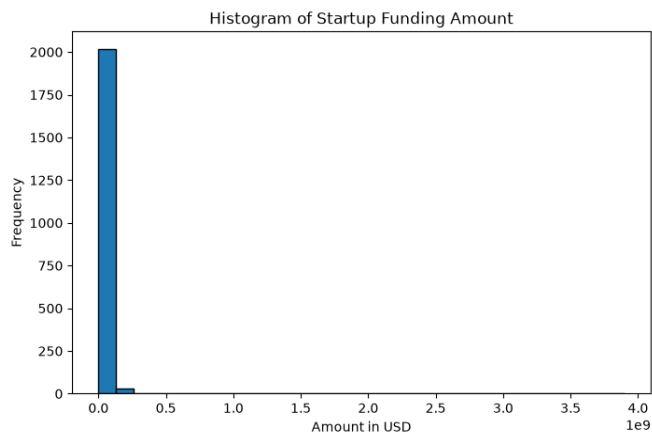
	Investors Name	InvestmentType	Amount in USD	Remarks
0	Tiger Global Management	Private Equity Round	20,00,00,000	NaN
1	Susquehanna Growth Equity	Series C	80,48,394	NaN
2	Sequoia Capital India	Series B	1,83,58,860	NaN
3	Vinod Khatumal	Pre-series A	30,00,000	NaN
4	Sprout Venture Partners	Seed Round	18,00,000	NaN

mean, median, mode,
variance

```
print("Mean:", funding.mean())  
print("Median:", funding.median())  
print("Mode:", funding.mode()[0])  
print("Variance:", funding.var())  
print("Standard Deviation:", funding.std())  
print("Minimum:", funding.min())  
print("Maximum:", funding.max())  
print("Range:", funding.max() - funding.min())
```

b. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

```
#2b  
import matplotlib.pyplot as plt  
plt.figure(figsize=(8,5))  
plt.hist(funding, bins=30, edgecolor='black')  
plt.title("Histogram of Startup Funding Amount")  
plt.xlabel("Amount in USD")  
plt.ylabel("Frequency")  
plt.show()
```



```
plt.figure(figsize=(8,4))  
plt.boxplot(funding, vert=False)  
plt.title("Box Plot of Startup Funding Amount")  
plt.xlabel("Amount in USD")  
plt.show()
```

